



RAMCO INSTITUTE OF TECHNOLOGY

Rajapalayam

Department of Computer Science and Engineering

Academic Year: 2019- 2020 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. Computer Science and Engineering.

Course Code & Title: IT6801 Service Oriented Architecture.

Name of the Faculty member: Dr.M.Kaliappan, Associate Professor/CSE

Theme of discussion: Web Services and SOA principles.

Date and Time: 18.09.2019 & 4.10 pm to 5.00

3 Innovative practices: JIGSAW Collaborative Learning

3.1 Topic:

Develop a web service application for implementing online Airline Reservation System using Service-Oriented Architecture principles and components. The system includes the following SOA components and principles.

1. **Web services & WSDL**
2. **SOAP**
3. **Message Exchange pattern**
4. **Web service Activity**
5. **Coordination service**
6. **Orchestration**

3.2 Learning outcomes (LO)

- LO1: To improve listening, communication, creativity and problem-solving skills
- LO2: To develop a simple web service application using SOA components and principles.

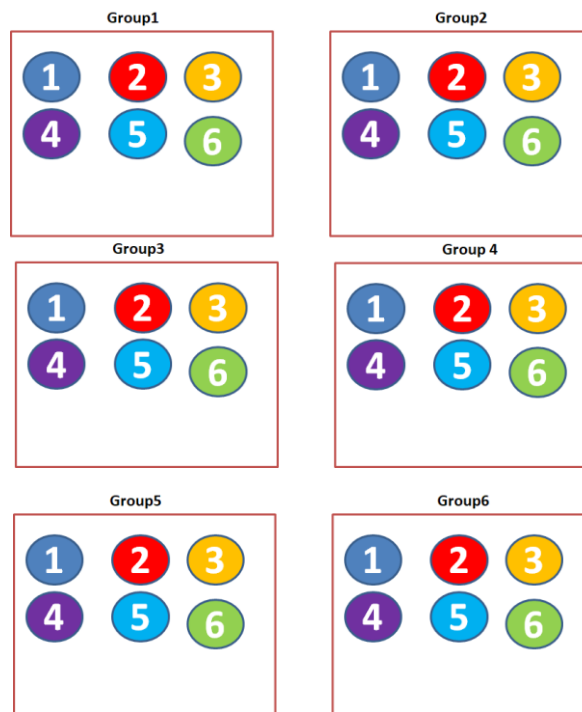
3.2 Justification for choosing the topic

This topic addresses all the core components of service-oriented principles. These core principles are applicable for designing any automation system. This topic is more suitable for a Jig-Saw cooperative learning activity in which students get exposure in all SOA components. Students can master all the core components of service-oriented principles. Students should be able to enhance their communication, peer learning, individual and teamwork, and self-learning by this activity.

3.3 Procedure and Implementation

Step 1 (5 minutes)

- Six groups with six members were formed based on the students' academic performance and ability. It is called home group.
- One student from each group selected as the leader. Each student in a group was given a number from 1 to 6. Each number is representing the following six SOA components. A student was assigned one SOA component in a home group.
 1. **Web services & WSDL**
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 6. **Orchestration**



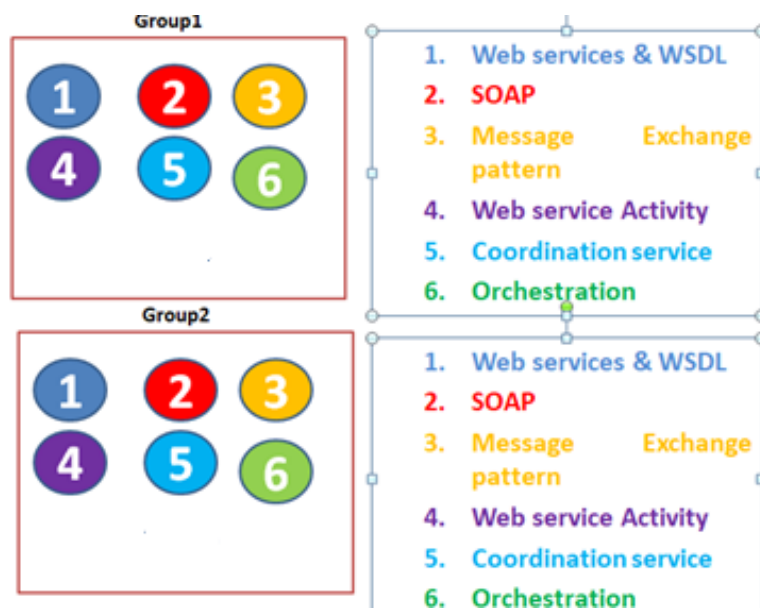
Step 2

- Expert groups were formed that consist of student from home groups who was assigned a same SOA component.
- Expert group members were given 15 minutes to discuss allotted SOA component.



Step 3 (15 minutes)

- The expert group students brought back into their home groups and allotted 15 minutes for further discussion in which each member become expert in the entire SOA component.



Step 4 (10 minutes)

- One student from each home group presented his learned content.
- The others in the group were encouraged to raise queries for clarification

Step 7(6 minutes)

- At the end of the session, conducted a quiz from the material.

Glimpse



3.5 Reflective Report

3.5.1 Challenges

- Participation of slow learner in the discussion
- Formation of heterogeneous group based on gender (male and female).

3.5.2 Observations:

The students were provided the opportunity to become experts in particular course content. They shared the knowledge they learned with peers. This learning ***promotes both self-learning and peer teaching*** among the students to understand the course content at a deeper level. Most of the students participated actively. The Jigsaw technique helped the students to develop expertise in a particular topic.

I motivated the students to form a heterogeneous group based on gender. Also, I had given a clue about the theme during the discussion for active participation. The students were encouraged to raise queries for clarification during the presentation.

3.5.3 Students Response:

- Most of the students discussed with teammates comfortably during brainstorming.
- The bright students were actively involved with peers to discuss the theme.
- The slow learners wanted to assist during the discussion. They were given tips about the theme of the discussion.
- Most of the students enjoyed the discussion in the expert group.

Students' response ensures that the students could improve listening, communication, creativity and problem-solving skills.

3.6 Relevance of program outcomes

Programme Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
LO1	3					2	2	2	2	2		2
LO2	3					2	2	2	2	2		2



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Feed Back

Innovative practice: Jigsaw cooperative learning technique.

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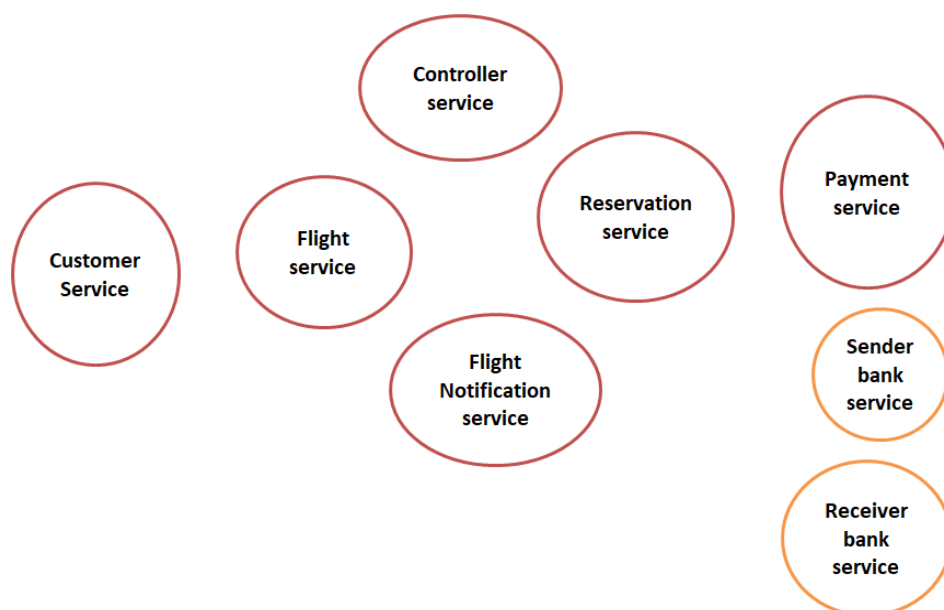
Theme of discussion: Web service application for implementing online Airline Reservation Systems.

Date and Time: 18.09.2019 & 4.10 pm to 5.00

Is encourage cooperative learning among yourself ?	Yes	No
Connection with course material in a creative and engaging way	Excellent	Good
Is improve listening, communication, and problem-solving skills?	Excellent	Good

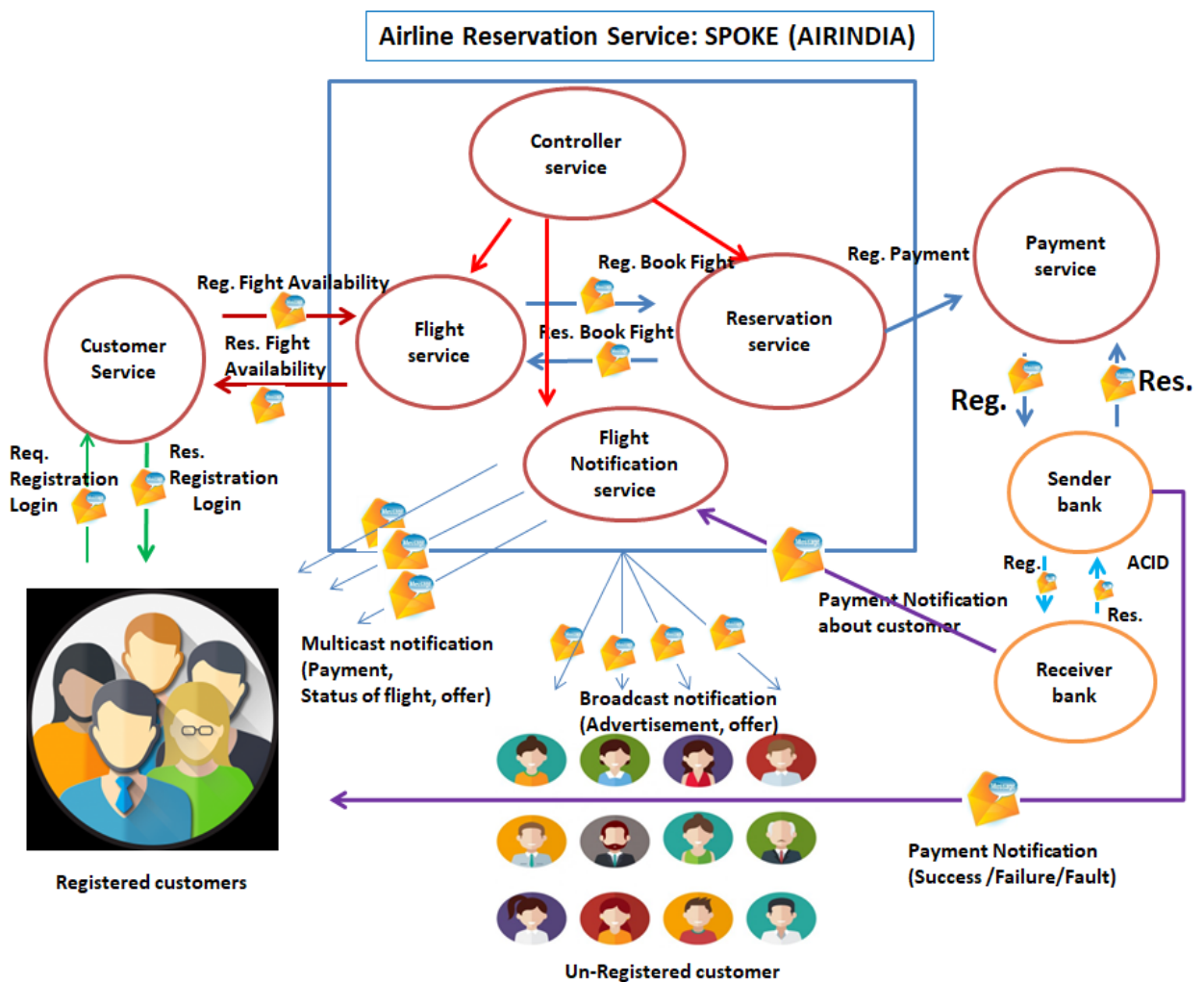
Result of the Activity:

1. Web services



2. Web services: Operations

- **Customer service : Operations**
 - Registration()
 - Login()
 - Update_customer_data()
- **Flight service : Operations**
 - Flight_availability()
 - Flight_information()
- **Reservation service : Operations**
 - Booking()
 - Cancel_booking()
 - Booking_status()
 - Generate_Booking()
- **Payment service : Operations**
 - Credit_card_validation()
 - Make_payment()
 - Check_balance()
 - Payment_Transaction()



1. WSDL:

Customer service - Login operation

```
<wsdl:portType name="CustomerServiceSoap">
  <wsdl:operation name="login">
    <wsdl:input message="tns:addSoapIn"/>
    <wsdl:output message="tns:addSoapOut"/>
  </wsdl:operation>
</wsdl:portType>
<wsdl:binding name="CustomerServiceSoap" type="tns:CustomerServiceSoap">
  <soap:binding transport="http://schemas.xmlsoap.org/soap/http"/>
  <wsdl:operation name="login">
    <soap:operation soapAction="http://tempuri.org/login" style="document"/>
    <wsdl:input>
      <soap:body use="literal"/>
    </wsdl:input>
    <wsdl:output>
      <soap:body use="literal"/>
    </wsdl:output>
  </wsdl:operation>
</wsdl:binding>
<wsdl:service name="CustomerService">
  <wsdl:port name="CustomerServiceSoap" binding="tns:CustomerServiceSoap">
    <soap:address location="http://localhost:61433/CustomerService.asmx"/>
  </wsdl:port>
</wsdl:service>
```

2.SOAP

Customer service - Login operation

SOAP input message(Service requestor)

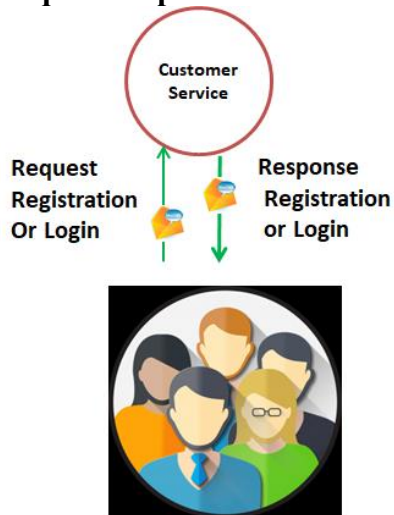
```
<?xml version="1.0" encoding="utf-8"?>
<soap:Envelope xmlns:xsi="http://www.w3.org"
  <soap:Body>
    <login xmlns="http://tempuri.org">
      <userid> string </userid>
      <password> string </password>
    </add>
  </soap:Body>
</soap:Envelope>
```

SOAP output message (Service provider)

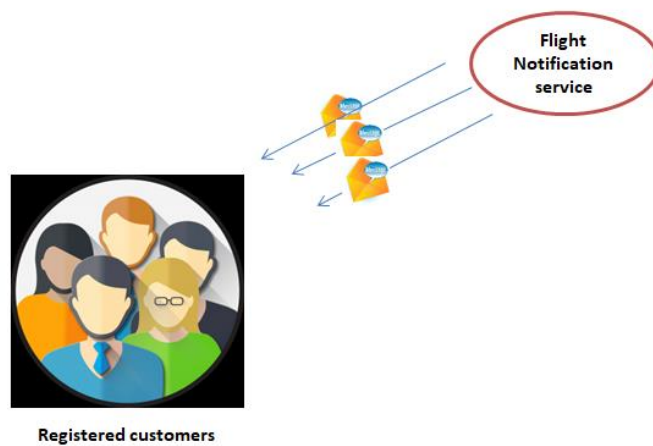
```
<?xml version="1.0" encoding="utf-8"?>
</soap:Envelope>
  <soap:Body>
    <addResponse xmlns="http://tempuri.org/">
      <addResult > boolean </addResult>
    </addResponse>
  </soap:Body>
</soap:Envelope>
```

3. Message Exchange pattern

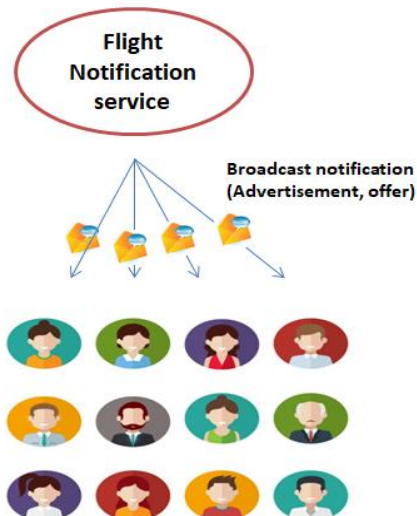
Request-response MEP



Multicast MEP



Broadcast MEP

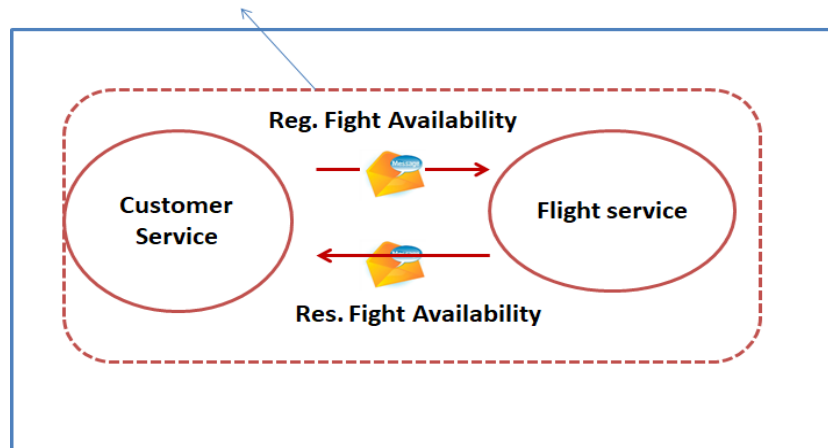


Notification or Out only MEP

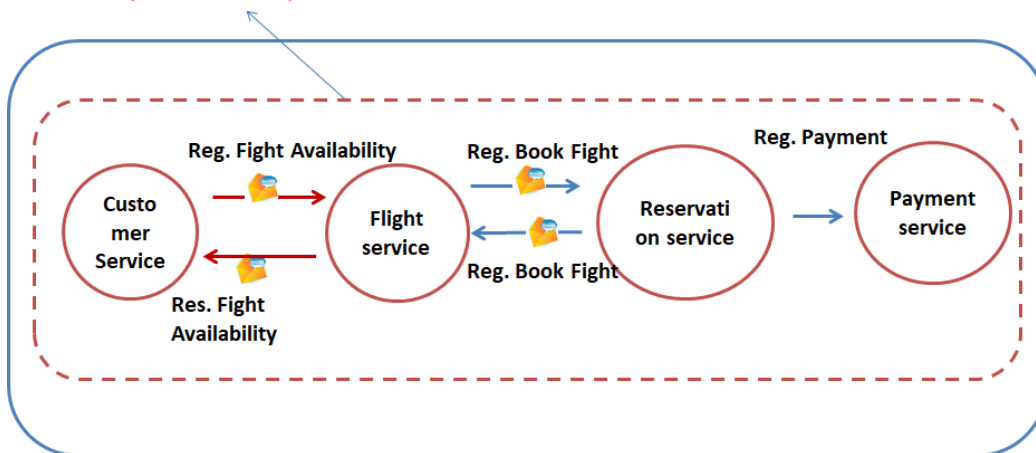


4. Web service Activity

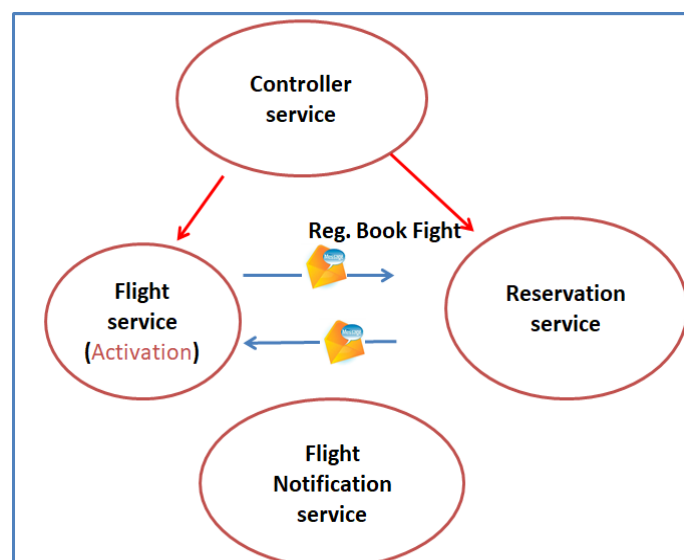
Primitive activity



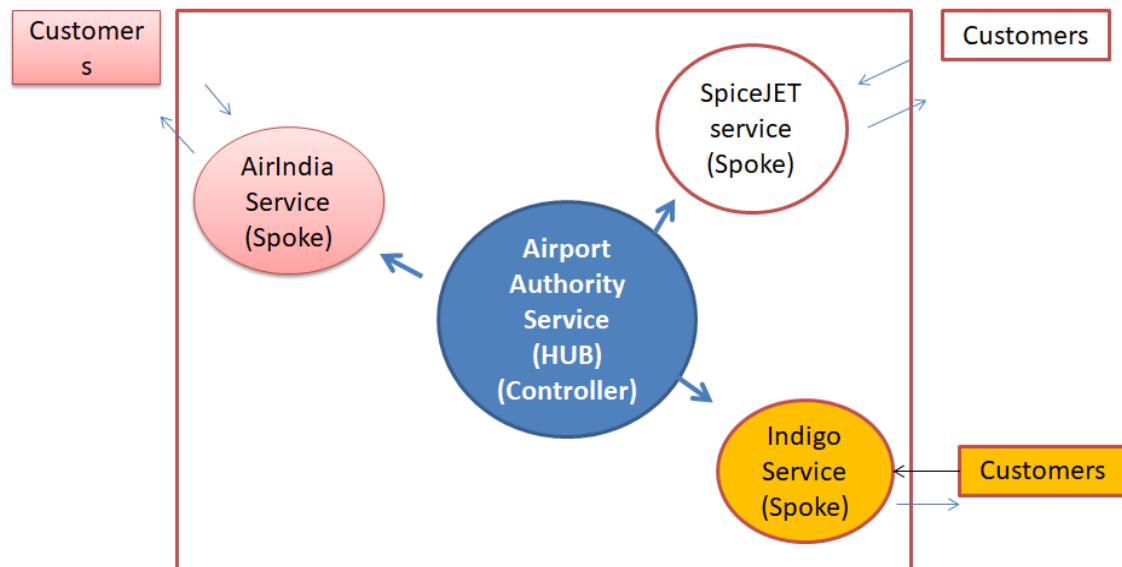
Complex activity



5. Coordination service



6. Orchestration



7. UDDI

```
<businessEntity businessKey = "uuid">
  operator = "www.airindia.com" authorizedName = "AICEO">
    <name> Air India Airline Reservation Company</name>
    <description> Airline Reservation Web services</description>
    <contacts>
      <contact useType = "general info">
        <personName>John </personName>
        <phone> (123) 123-1234</phone>
        <email> john@airindia.com</email>
      </contact>
    </contacts>
  </businessEntity>

  <businessService serviceKey = "uuid1" businessKey = "uuid:A">
    <name> Controller services</name> <description> ....</description>
  </businessService>
  <businessService serviceKey = "uuid2" businessKey = "uuid:B">
    <name> Customer services</name> <description> ....</description>
  </businessService>
  <businessService serviceKey = "uuid3" businessKey = "uuid:C">
    <name> Flight services</name> <description> ....</description>
```

```

</businessService>
<businessService serviceKey = "uuid4" businessKey = "uuid:D">
    <name> Reservation services</name> <description> ....</description>
</businessService>
<businessService serviceKey = "uuid5" businessKey = "uuid:E">
    <name> payment services</name> <description> ....</description>
</businessService>

<bindingTemplate serviceKey = "uuid1" bindingKey = "uuid:C001">
    <description> ControllerSOAP Binding</description>
    <accessPoint URLType = "http"> http://localhost:8080 </accessPoint>
</bindingTemplate>
<bindingTemplate serviceKey = "uuid2" bindingKey = "uuid:C002">
    <description> CustomerSOAP Binding</description>
    <accessPoint URLType = "http"> http://localhost:8080 </accessPoint>
</bindingTemplate>
<bindingTemplate serviceKey = "uuid3" bindingKey = "uuid:C003">
    <description> FlightSOAP Binding</description>
    <accessPoint URLType = "http"> http://localhost:8080 </accessPoint>
</bindingTemplate>
<tModel tModelKey = "uuid:xyz987.." operator = "www.airindia.com" authorizedName =
"John">
    <name> AirlineReservation Interface Port Type</name>

<overviewDoc>
    <overviewURL> http://localhost/ AirlineReservation.wsdl
    </overviewURL>
</overviewDoc>
</tModel>

```

Reference:

1. <https://www.jigsaw.org/>